Technical Companion:

The Solved Theorem Stack of Adequacy Architecture

Program 1: Finite Admissibility, Pullback Unification,
Grammar-Cut Maximality, and the NF4 Boundary

Adequacy Architecture Project

2026-03-27 (machine-checked Lean 4, 0 `sorry`)

Companion to: *The Architecture of Outer Admissibility* (Adequacy_Flagship_Paper.tex)

Abstract

We present a machine-checked layered certificate architecture for outer admissibility in the adequacy framework. The central result is that the finite admissibility gate G_{adm} is logically equivalent to the pullback display witness U_{pullback} for all type-indexed slots γ (*Program 1 Unification*). The unified gate implies the existence of an absolute frontier S1 certificate and is maximal under the declared OG1 grammar (normal forms NF0–NF3). The nearest boundary candidate—the NF4 win predicate S_1^{lawful}—is Layer-B sound but does not force grammar expansion (*NF4-C verdict*). Above the gate, a functorial second layer (L_2, the S5 / summit-FE3 package) is canonically produced from any admission witness without additional hypotheses. Stronger Stage-D content is residual: IC-specific, not implied by U_{pullback} alone, and partially reconstructible from named hooks. The resulting picture is a strict three-level architecture—outer gate, functorial second layer, residual band—every boundary of which is characterized by a named, machine-checked theorem. All proofs are in Lean 4 with 0 `sorry`; exact names and modules are given in the appendix.

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1 Introduction

The adequacy architecture defines a class of outer certificate predicates $C : \mathsf{Type} \to \mathsf{Prop}$ evaluated at a type-indexed slot γ. The fundamental question is: *which predicates C are genuinely admissible*—meaning they imply the existence of an absolute frontier S1 representation?

This is not a trivial question. Naively, any predicate that happens to hold at some γ could be called a "certificate." Program 1 disciplines this by defining a finite, constructively presented family of admissible outer modes and proving that all of them collapse to a single unified gate.

The main results of the program, all machine-checked, are:

(I) **Unification.** $G_{\mathrm{adm}}\,\gamma \iff U_{\mathrm{pullback}}\,\gamma$ for all γ.

(II) **Flagship consequence.** $G_{\mathrm{adm}}\,\gamma \Rightarrow \exists P\,B$, $\mathrm{AbsRawS1}(P, B)$.

(III) **Grammar-cut maximality.** Every grammar-generated outer class (OG1, NF0–NF3) is contained in U_{pullback}.

(IV) **NF4-C boundary.** The nearest expansion candidate (S_1^{lawful}) is Layer-B sound but not universally reducible to U_{pullback}; the grammar boundary is stable.

(V) **Second layer.** $U_{\mathrm{pullback}}\,\gamma \Rightarrow L_2\,\gamma$ (S5 / summit-FE3 pullback-display package), a functorial consequence, not a new outer mode.

(VI) **Stage D residuality (SR-B).** $\mathrm{StageD}\,\gamma$ is not implied by $U_{\mathrm{pullback}}\,\gamma$; it carries a genuine IC-local residue.

The paper is organized as follows. Section 2 establishes the setting and notation. Sections 3–6 present the three layers. Section 4 discusses the NF4 boundary. Section 7 synthesizes the full architectural picture. Section 8 states the explicitly open frontiers. The appendix (§A) gives exact Lean theorem names, modules, and the dependency graph.

2 Setting and notation

2.1 The outer certificate world

Fix a universe of types. An *outer certificate predicate* is a predicate $C : \mathsf{Type} \to \mathsf{Prop}$. We evaluate such predicates at a type-indexed slot γ. The slot γ is the outer variable throughout Program 1.

Definition 2.1 (Finite admissibility gate)**.** $G_{\mathrm{adm}}\,\gamma$ (`Program1FiniteGAdm`) holds when γ is witnessed by one of three constructor arms:

1. A *pullback display arm*: a `Program1AdmissibilityPullbackDisplayWitness` γ.
2. An *IC Stage-D full pack arm*: the full `icSpec031_stageD_resolution_pack` at γ.
3. An *upgrade witness arm*: a `Program1CertifiedUpgradeWitness` γ.

Definition 2.2 (Pullback display witness)**.** $U_{\mathrm{pullback}}\,\gamma$ (`Program1AdmissibilityPullbackDisplayWitness`) holds when there is a `CertifiedFrontierRepresentation` (or equivalent display witness) obtained by pulling back a lawful adequacy architecture along a compare map at γ.

U_{pullback} is not vacuous: the IC case threads through a compression carrier pullback proved in `SPEC_031_ZK9`; the corpus swap case uses an involutive compare-map cleaving proved in `SPEC_035_UC1`.

2.2 Notation summary

Symbol	Lean name	Meaning
$G_{\mathrm{adm}}\,\gamma$	`Program1FiniteGAdm`	finite admissibility gate
$U_{\mathrm{pullback}}\,\gamma$	`Program1AdmissibilityPullbackDisplayWitness`	pullback display witness
G_{gen}	`Program1GrammarGeneratedOuterGate`	OG1 grammar-generated outer gate
$\mathrm{AbsRawS1}(P, B)$	`AbsoluteFrontierRawS1`	absolute frontier S1 certificate
$S_1^{\mathrm{lawful}}\,\gamma$	`Nonempty (S1LawfulFrontierRow γ)`	NF4 win predicate
$L_2\,\gamma$	`pullbackDisplay_with_host_summitFE3`	S5 / summit-FE3 layer-2 package
$\mathrm{StageD}\,\gamma$	`icSpec031_stageD_resolution_pack`	IC Stage-D four-conjunct pack

3 Layer A: The outer gate theorem

3.1 Unification

Theorem 3.1 (Program 1 Unification)**.** *For all* γ, $\quad G_{\mathrm{adm}}\,\gamma \iff U_{\mathrm{pullback}}\,\gamma$.

```
program1FiniteGAdm_iff_program1AdmissibilityPullbackDisplayWitness :
  forall gamma, G_adm gamma <-> U_pullback gamma
-- Module: Instances/Program1MetaUnificationMU1.lean
```

Proof sketch. Each of the three constructor arms of G_{adm} is shown to produce a U_{pullback} witness, and conversely. The IC arm uses the compression carrier pullback from `SPEC_031_ZK9`. The upgrade arm factors through `CertifiedSummitRowRepresentation` via the representation-calculus

gate in SPEC_033_RL9. The pullback arm is definitionally the first constructor and the conversion is immediate. The converse direction packages any U_{pullback} witness into the first constructor arm of G_{adm}. □

Remark 3.2. The nontriviality runs in both directions. From above: three structurally different arms collapse to one. From below: U_{pullback} requires a genuine lawful adequacy architecture pullback; it is not trivially satisfied.

3.2 Flagship consequence

Theorem 3.3 (Absolute frontier S1 from admission)**.** *For all* γ, $\quad G_{\text{adm}}\,\gamma \Rightarrow \exists P\,B,\ \text{AbsRawS1}(P, B)$.

```
program1FiniteGAdm_implies_exists_absoluteFrontierRawS1 :
  forall gamma, G_adm gamma -> exists P B, AbsoluteFrontierRawS1 P B
-- Module: Instances/Program1MetaUnificationMU1.lean
```

This is the primary outer payoff: admission at any slot witnesses the non-emptiness of the absolute frontier.

3.3 Grammar-cut maximality

Theorem 3.4 (Grammar-cut maximality)**.** *Let* $C : \mathsf{Type} \to \mathsf{Prop}$. *If* C *is generated by the OG1 grammar* $(G_{\text{gen}}\,C)$, *then* $\forall\,\gamma,\ C\,\gamma \Rightarrow U_{\text{pullback}}\,\gamma$.

```
program1_outer_grammar_cut_maxA :
  forall (C : Type -> Prop),
    Program1GrammarGeneratedOuterGate C ->
    forall gamma, C gamma -> U_pullback gamma
-- Module: Instances/Program1OuterGateGrammar.lean
```

Proof sketch. The OG1 grammar is the closure of the normal forms NF0 (bare pullback display), NF1 (certified frontier row), NF2 (upgrade witness), NF3 (composition) under the declared grammar rules. Each normal form is shown to produce a U_{pullback} witness; the closure induction is straightforward. □

Theorem 3.4 is the correct form of a maximality claim: it does not assert maximality over all possible outer predicates (that would be false; see §4), but over the disciplined grammar. Within that scope, U_{pullback} is the ceiling.

4 Boundary analysis: the NF4 finding

4.1 What NF4 is

Define the *NF4 win predicate* $S_1^{\text{lawful}}\,\gamma := \mathsf{Nonempty}(\texttt{S1LawfulFrontierRow}\,\gamma)$. NF4 is the nearest principled candidate for a broader outer class: it lies outside the OG1 grammar, but is related to the gate by a soundness result.

Theorem 4.1 (NF4 Layer-B soundness)**.** *For all* γ, $\quad S_1^{\text{lawful}}\,\gamma \Rightarrow$ *program1OuterLayerBLaw*γ.

```
Program1OuterCertificateSound_program1OuterWinS1LawfulNonempty :
  forall gamma, S1Lawful gamma -> program1OuterLayerBLaw gamma
-- Module: Instances/Program1MaximalityMA1.lean
```

4.2 The NF4-C verdict

The NF4-C verdict records three facts jointly:

1. S_1^{lawful} is Layer-B sound (Theorem 4.1).
2. S_1^{lawful} is *not* grammar-generated under OG1 (not in NF0–NF3).
3. There is *no proved universal* $S_1^{\text{lawful}}\,\gamma \Rightarrow U_{\text{pullback}}\,\gamma$.

NF4 therefore remains a *logical pocket*: sound where instantiated, but not promoted to the operative outer gate.

Remark 4.2. NF4-C strengthens Theorem 3.4: the grammar boundary is not an artifact of under-specification. The nearest escape hatch was examined and did not force expansion. The architecture is stable at its current boundary.

5 Layer B: The functorial second layer

Theorem 5.1 (S5 functorial second layer)**.**

$$\begin{aligned} &\forall\,\gamma, \quad U_{\text{pullback}}\,\gamma \;\Rightarrow\; L_2\,\gamma \\ &\forall\,\gamma, \quad G_{\text{adm}}\,\gamma \;\Rightarrow\; L_2\,\gamma \end{aligned}$$

```
program1_layer2PullbackDisplay_with_host_summitFE3_of_U_pullback :
  forall gamma, U_pullback gamma -> L2 gamma
program1_layer2PullbackDisplay_with_host_summitFE3_of_program1FiniteGAdm :
  forall gamma, G_adm gamma -> L2 gamma
-- Module: Instances/Program1LayeredStructureLS1.lean
```

Proof sketch. The L_2 package (`pullbackDisplay_with_host_summitFE3`) is obtained by projecting the summit-FE3 witness from the same display structure that witnesses $U_{\text{pullback}}\,\gamma$. No additional hypotheses are required: the display witness contains all the data needed to produce the host-side `summitFE3` certificate. □

The structural point is that L_2 is not a new constructor of G_{adm}. It is not a new way to be admitted; it is what admission *unlocks*. Specifically:

- $L_2\,\gamma$ is not grammar-generated (not in NF0–NF3, not in NF4).
- $L_2\,\gamma$ does not imply $G_{\text{adm}}\,\gamma$ (it is downstream).
- $L_2\,\gamma$ is the unique canonically produced second-layer structure from any admission witness.

The outer certificate world therefore has a strict two-level structure: Level 1 is admission; Level 2 is the functorial L_2 consequence.

6 Layer C: Stage D as residual content

6.1 The Stage D pack

Stage D is the IC-native condition connecting the IC compression architecture to the corpus H_{final}. It has a four-conjunct resolution pack:

Theorem 6.1 (Stage D resolution)**.** *For all γ,*

$$\text{StageD}\,\gamma \iff F_1(\gamma) \wedge F_2(\gamma) \wedge F_3(\gamma) \wedge F_4(\gamma)$$

where F_1 is the AF1 obstruction conjunct (`af1_obstruction_conjunct`), F_2 is the absolute-frontier S1 conjunct (`absoluteFrontierRawS1_conjunct`), F_3 is the bypass conjunct (`bypass_conjunct`), and F_4 is the cascade conjunct (`cascade_conjunct`).

```
icSpec031_stageD_resolution_pack :
  forall gamma, StageD gamma <->
    af1_obstruction_conjunct gamma /\
    absoluteFrontierRawS1_conjunct gamma /\
    bypass_conjunct gamma /\
    cascade_conjunct gamma
-- Module: Instances/ProgramFiniteAdmissibility.lean
```

Named facet hooks (SR1 reconstruction, from `SPEC_043`):

```
icSpec031_stageD_iff_conjunct_of_af1_native_carrier  -- F1: iff spine
icSpec031_stageD_bypass_conjunct_of_af1_standard     -- F3: bypass
icSpec031_stageD_cascade_conjunct_of_af1_standard    -- F4: cascade
```

6.2 SR-B residuality verdict

Proposition 6.2 (Stage D residuality)**.** *(a) $U_{\text{pullback}}\,\gamma$ does not imply* StageD γ *in general.*

(b) StageD γ *does not imply $U_{\text{pullback}}\,\gamma$ without the F_1 and F_3/F_4 arms.*

(c) F_1, F_3, F_4 are reconstructible from named hypotheses (partial basis).

(d) F_2 (the AbsRawS1 *conjunct) is the irreducible residue: it does not follow from $U_{\text{pullback}}\,\gamma$ alone.*

Stage D is therefore *residual extra structure*: it sits above the outer gate and the S5 layer, is partially accessible via named Lean hooks, but carries a genuine IC-local obstruction that is not reached by U_{pullback} alone. This is the SR-B verdict from `SPEC_043_SR1`.

7 The full architecture

The six results of §§3–6 combine into a single coherent picture. Every boundary is characterized by what does and does not follow from U_{pullback} alone.

7.1 The three-level diagram

```
NF4 (S1Lawful)  -- Layer-B sound; not grammar-generated;
                   not universally = U_pullback  [NF4-C]
     ^
-------------------------------------------------------
Residual: Stage D  (IC-local; SR-B; partial basis above
                    U_pullback; irreducible F2)
     ^
-------------------------------------------------------
Level 2: L2  (pullbackDisplay_with_host_summitFE3)
             [functorial from U_pullback; not an outer mode]
     ^
-------------------------------------------------------
Level 1: G_adm <=> U_pullback  (outer gate; maximal NF0--NF3)
     v
exists P B, AbsoluteFrontierRawS1 P B
```

Every arrow is a named theorem. The horizontal rules are not informal "levels" — they are precise logical boundaries characterized by what is and is not derivable from U_{pullback} alone.

7.2 The integrated claim

For the disciplined Program-1 certificate world:

(1) **Unique unified outer gate.** $G_{\text{adm}} \equiv U_{\text{pullback}}$ (Theorem 3.1).

(2) **Absolute frontier consequence.** $G_{\text{adm}}\,\gamma \Rightarrow \exists P\,B$, $\text{AbsRawS1}(P, B)$ (Theorem 3.3).

(3) **Grammar maximality.** No OG1-grammar-generated class exceeds U_{pullback} (Theorem 3.4).

(4) **Stable boundary.** The nearest expansion candidate (NF4) does not force grammar expansion (NF4-C, §4).

(5) **Functorial second layer.** $U_{\text{pullback}}\,\gamma \Rightarrow L_2\,\gamma$, canonically, without additional hypotheses (Theorem 5.1).

(6) **Residual band.** StageD is not primitive to the outer gate; it carries a genuine IC-local residue (Proposition 6.2).

7.3 What the architecture is not claiming

- It does not claim maximality over *all* possible outer predicates. NF4 exists and is not grammar-generated; the maximality result is correctly scoped to the OG1 grammar.
- It does not claim that AbsRawS1 holds for arbitrary (P, B) without hypotheses. Global raw $\forall P\,B$ is the honest post-closure frontier per `SPEC_034` and remains open.
- It does not claim that Stage D is unreachable or unimportant. SR-B says it has a partial basis and an irreducible residue; it does not close off future work on the IC-local spine.

8 Open frontiers

Three frontiers are explicitly left open:

(I) **Global raw $\forall P\, B$.** AbsRawS1(P, B) for an arbitrary predicate pair without hypotheses. This is the honest post-closure frontier per `SPEC_034_PC7`.

(II) **Broader outer grammar.** If a principled broader-than-OG1 grammar is chartered in a future program, the grammar-cut maximality result would need re-examination in that scope. NF4-C is not a proof that no broader grammar is possible; it is a proof that the nearest candidate did not force one now.

(III) **Stage D universality.** Whether StageD could ever be brought inside the outer gate for all γ, or whether the IC-local obstruction (F2) is genuinely permanent. The SR-B verdict gives "partial basis, irreducible residue," and does not close off future work.

These are not gaps in the current theorem stack. They are the next genuine frontiers, now clearly located by the architecture.

Acknowledgements

All proofs are machine-checked in Lean 4 (toolchain `leanprover/lean4:v4.29.0-rc6`, Mathlib `v4.29.0-rc6`). Reproducibility: run `lake build` from `adequacy-architecture-lean/` and inspect the named theorems in the modules listed in §A.

A Lean theorem names and modules

Complete name index: `THEOREM_INVENTORY.md` (rows SPEC_036–SPEC_044).

Theorem	Lean name	Module
3.1	`program1FiniteGAdm_iff_...PullbackDisplayWitness`	*Program1MetaUnificationMU1.l*
3.3	`program1FiniteGAdm_implies_exists_absoluteFrontierRawS1`	*Program1MetaUnificationMU1.l*
3.4	`program1_outer_grammar_cut_maxA`	*Program1OuterGateGrammar.le*
4.1	`Program1OuterCertificateSound_program1OuterWinS1LawfulNonempty`	*Program1MaximalityMA1.lean*
5.1	`program1_layer2PullbackDisplay_with_host_summitFE3_of_U_pullback`	*Program1LayeredStructureLS1.le*
5.1	`program1_layer2PullbackDisplay_with_host_summitFE3_of_...GAdm`	*Program1LayeredStructureLS1.le*
6.1	`icSpec031_stageD_resolution_pack`	*ProgramFiniteAdmissibility.lean*
SR1 F1	`icSpec031_stageD_iff_conjunct_of_af1_native_carrier`	*ProgramFiniteAdmissibility.lean*
SR1 F3	`icSpec031_stageD_bypass_conjunct_of_af1_standard`	*ProgramFiniteAdmissibility.lean*
SR1 F4	`icSpec031_stageD_cascade_conjunct_of_af1_standard`	*ProgramFiniteAdmissibility.lean*

Module paths

All modules are under `adequacy-architecture-lean/AdequacyArchitecture/`.

- *Instances/Program1MetaUnificationMU1.lean* — SPEC_037 / outer unification
- *Instances/Program1OuterGateGrammar.lean* — SPEC_041 OG1 grammar

- *Instances/Program1MaximalityMA1.lean* — SPEC_040 / SPEC_044 maximality + NF4
- *Instances/Program1LayeredStructureLS1.lean* — SPEC_042 LS1 second layer
- *Instances/ProgramFiniteAdmissibility.lean* — SPEC_036 / SPEC_043 Stage D

Source of record

Mathematical kernel: `specs/IN-PROCESS/SPEC_045_AS1/SPEC_045_AS1_A_INTEGRATED_THEOREM_STATEMENT.md`
Structural explanation: `specs/IN-PROCESS/SPEC_045_AS1/SPEC_045_AS1_B_STRUCTURAL_EXPLANATION.md`
Artifact map: `specs/IN-PROCESS/SPEC_045_AS1/SPEC_045_AS1_C_ARTIFACT_AND_THEOREM_MAP.md`
Reader thread: `specs/IN-PROCESS/SPEC_045_AS1/SPEC_045_AS1_D_FLAGSHIP_NARRATIVE.md`